Math for CS & AI: Homework 2

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Problem 3.8.8

Prove the following identities:

$$\sum_{k=0}^{m}(-1)^k\binom{n}{k} = (-1)^m\binom{n-1}{m}; \tag{1}$$

$$\sum_{k=0}^{n}\binom{n}{k}\binom{k}{m} = \binom{n}{m}2^{n-m}. \tag{2}$$

Answer:

For equation (1),

$$\begin{aligned}\sum_{k=0}^{m}(-1)^k\binom{n}{k} &= \sum_{k=0}^{m}(-1)^k\binom{n-1}{k} + \sum_{k=0}^{m}(-1)^k\binom{n-1}{k-1} \\ &= (-1)^m\binom{n-1}{m} + \sum_{k=0}^{m-1}\binom{n-1}{k}\left[(-1)^k + (-1)^{k+1}\right] \\ &= (-1)^m\binom{n-1}{m}.\end{aligned}$$

For equation (2),

$$\begin{aligned}\sum_{k=0}^{n}\binom{n}{k}\binom{k}{m} &= \sum_{k=m}^{n}\frac{n!}{k!(n-k)!}\cdot\frac{k!}{m!(k-m)!} \\ &= \sum_{k=m}^{n}\frac{n!}{m!(n-m)!}\cdot\frac{(n-m)!}{(n-k)!(k-m)!} \\ &= \binom{n}{m}\sum_{k=0}^{n-m}\binom{n-m}{k} \\ &= \binom{n}{m}2^{n-m}.\end{aligned}$$

Problem 3.8.14

Let n be a positive integer divisible by 3. Use Stirling's formula to find the approximate value of $\binom{n}{n/3}$.

Answer:

Applying the Stirling's formula, which states that $n! \sim \left(\frac{n}{e}\right)^n \sqrt{2\pi n}$, we obtain:

$$\begin{aligned}
\binom{n}{n/3} &= \frac{n!}{(2n/3)!(n/3)!} \\
&\sim \frac{\left(\frac{n}{e}\right)^n \sqrt{2\pi n}}{\left(\frac{2n}{3e}\right)^{2n/3} \sqrt{\frac{4\pi n}{3}} \cdot \left(\frac{n}{3e}\right)^{n/3} \sqrt{\frac{2\pi n}{3}}} \\
&= \frac{3^n}{2^{2n/3}} \cdot \sqrt{\frac{9}{4\pi n}} \\
&= \left(\frac{27}{4}\right)^{n/3} \cdot \frac{3}{2}\sqrt{\frac{1}{\pi n}}.
\end{aligned}$$

Therefore, $\binom{n}{n/3} \sim \left(\frac{27}{4}\right)^{n/3} \cdot \frac{3}{2}\sqrt{\frac{1}{\pi n}}$.

Problem 10

In a New Year's party with $2n$ people, k random names are picked to receive gifts. Assume that these $2n$ people are actually n couples. Let $r_{n,k}$ be the probability that there exists at least one couple such that both people in the couple receive gifts.

(a) Express $r_{n,k}$ as a mathematical formula of n and k.

(b) Determine the value of $\lim_{n\to\infty} r_{n,\lfloor\sqrt{n}\rfloor}$.

Answer:

The probability $r_{n,k}$ can be expressed by using its complement event as follows:

$$r_{n,k} = 1 - \frac{\binom{n}{k}2^k}{\binom{2n}{k}}.$$

In this formula, $\binom{2n}{k}$ represents the total number of possible selections, while $\binom{n}{k}2^k$ represents the number of selections where no couple is entirely chosen, which is achieved by selecting k couples and choosing one person from each selected couple.

Expanding the combination terms, we obtain:

$$\begin{aligned}
\overline{r_{n,k}} &= \frac{\binom{n}{k}2^k}{\binom{2n}{k}} \\
&= \prod_{i=0}^{k-1} \frac{2n-2i}{2n-i} \\
&= \exp\left[\sum_{i=0}^{k-1} \ln\left(1-\frac{i}{n}\right) - \ln\left(1-\frac{i}{2n}\right)\right].
\end{aligned}$$

Now, we compute the limit of $\overline{r_{n,k}}$ as $n \to \infty$ with $k = \lfloor\sqrt{n}\rfloor$:

$$\begin{aligned}
\lim_{n\to\infty} \overline{r_{n,k}} &= \exp\left[\sum_{i=0}^{k-1} \ln\left(1-\frac{i}{n}\right) - \ln\left(1-\frac{i}{2n}\right)\right] \\
&= \exp\left[\sum_{i=0}^{k-1} -\frac{i}{2n} + \mathcal{O}\left(\frac{i^2}{n^2}\right)\right] \\
&= \exp\left(-\frac{k(k-1)}{4n}\right) \cdot \exp\left(\mathcal{O}\left(\frac{k^3}{n^2}\right)\right) \\
&= e^{-1/4},
\end{aligned}$$

where $k = \lfloor\sqrt{n}\rfloor$ is used. Therefore, $\lim_{n\to\infty} r_{n,k} = 1 - e^{-1/4}$.

Problem 11

Prove Statement 2 of Theorem 2.45: the optimal threshold satisfies $k^* = n/e + \mathcal{O}(1)$.

Answer:

Recall that $p_{n,k} = \frac{k}{n}(H_{n-1} - H_{k-1})$, which implies:

$$\begin{aligned} p_{n,k+1} - p_{n,k} &= \frac{k+1}{n}(H_{n-1} - H_k) - \frac{k}{n}(H_{n-1} - H_{k-1}) \\ &= \frac{1}{n}(H_{n-1} - H_k - 1). \end{aligned}$$

Since this expression is monotonically decreasing, the sequence $\{p_{n,k}\}_{1 \le k \le n-1}$ has a single peak.

Using the asymptotic expansion of harmonic numbers, we have:

$$H_{n-1} - H_k - 1 = \ln \frac{n}{ke} + \mathcal{O}\left(\frac{1}{n}\right).$$

The peak k^* is the smallest integer k such that $p_{n,k+1} - p_{n,k} \le 0$, which leads to the condition:

$$0 \le |H_{n-1} - H_{k^*} - 1| < \frac{1}{k^*}.$$

From this, we derive:

$$\begin{aligned} \ln \frac{n}{k^* e} &= \mathcal{O}\left(\frac{1}{n}\right), \\ \frac{n}{k^* e} &= \exp\left(\mathcal{O}\left(\frac{1}{n}\right)\right), \\ \frac{n}{k^* e} &= 1 + \mathcal{O}\left(\frac{1}{n}\right), \\ k^* &= \frac{n}{e} \cdot \frac{1}{1 + \mathcal{O}\left(\frac{1}{n}\right)}, \\ k^* &= \frac{n}{e} \cdot \left(1 + \mathcal{O}\left(\frac{1}{n}\right)\right), \\ k^* &= \frac{n}{e} + \mathcal{O}(1). \end{aligned}$$

where we used Taylor expansions $\exp x = 1 + x + \mathcal{O}(x^2)$ and $(1+x)^\alpha = 1 + \alpha x + \mathcal{O}(x^2)$ as $x = \mathcal{O}\left(\frac{1}{n}\right)$.

Problem 12

For the locker puzzle, let h_n be the success probability of the pointer chasing strategy when every student is allowed to open at most $n/3$ lockers n is a multiple of 3). Determine the value of $\lim_{n\to\infty} h_n$.

Answer:

Similarly, the pointer chasing strategy succeeds if and only if every cycle has a length of at most $n/3$. Therefore, the failure event $\overline{A}$ is:

$$\overline{A} = \{\sigma \in \Omega : \sigma \text{ has a cycle of length } j > n/3\}.$$

To account for the possibility of multiple cycles exceeding the length $n/3$, we apply the inclusion-exclusion principle to compute $\left|\overline{A}\right|$:

$$\left|\overline{A}\right| = \sum_{j=n/3+1}^{n} \binom{n}{j}(j-1)!(n-j)! - \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\sum_{k=n/3+1}^{n-j}\binom{n}{j+k}\binom{j+k}{j}(j-1)!(k-1)!(n-j-k)!.$$

Each scheme with two invalid cycles is counted twice in the first sum and subtracted once in the second sum, ensuring it is only counted once overall. We then express the probability of the failure event as:

$$\begin{aligned}
\Pr[\overline{A}] &= \frac{|\overline{A}|}{|\Omega|} \\
&= \sum_{j=n/3+1}^{n}\frac{1}{j} - \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\sum_{k=n/3+1}^{n-j}\frac{1}{jk} \\
&= H_n - H_{n/3} - \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\frac{H_{n-j}-H_{n/3}}{j} \\
&= \ln 3 - \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\frac{1}{j}\left[\ln 3 + \ln\left(1-\frac{j}{n}\right) + \mathcal{O}\left(\frac{1}{n}\right)\right] + \mathcal{O}\left(\frac{1}{n}\right) \\
&= \ln 3 \cdot \left(1 - \frac{H_{2n/3}-H_{n/3}}{2}\right) - \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\frac{1}{j}\ln\left(1-\frac{j}{n}\right) + \mathcal{O}\left(\frac{1}{n}\right) \\
&= \ln 3 \cdot \left(1 - \frac{\ln 2}{2}\right) - \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\frac{1}{j}\ln\left(1-\frac{j}{n}\right) + \mathcal{O}\left(\frac{1}{n}\right).
\end{aligned}$$

Therefore, the probability of success event A is:

$$\begin{aligned}
\Pr[A] &= 1 - \Pr[\overline{A}] \\
&= 1 - \ln 3 \cdot \left(1 - \frac{\ln 2}{2}\right) + \frac{1}{2}\sum_{j=n/3+1}^{2n/3}\frac{1}{j}\ln\left(1-\frac{j}{n}\right) + \mathcal{O}\left(\frac{1}{n}\right) \\
&\approx 4.86\%.
\end{aligned}$$

Problem 13

Let p be a prime number. In number theory, one definition of a prime number is as follows: a positive integer $p > 1$ is called a prime number if for all integers a and b, if $p \mid ab$, then $p \mid a$ or $p \mid b$. Here $p \mid a$ means that p divides a (or equivalently, a is a multiple of p).

Prove the following statements for all $a \in \mathbb{Z}_p \setminus \{0\} = \{1, 2, \ldots, p-1\}$:

(a) The congruence

$$ax \equiv 0 \pmod{p},$$

has exactly one solution x among $\{0, 1, \ldots, p-1\}$, which is $x = 0$.

(b) The following set equality holds:

$$\{ax \bmod p : x \in \mathbb{Z}_p \setminus \{0\}\} = \mathbb{Z}_p \setminus \{0\}.$$

(c) The following congruence holds, known as Fermat's little theorem:

$$a^{p-1} \equiv 1 \pmod{p}.$$

(d) There exists a unique $b \in \mathbb{Z}_p \setminus \{0\}$ such that $ab \equiv 1 \pmod{p}$.

Answer:

For statement (a), since $ax \equiv 0 \pmod{p}$ is equivalent to $p \mid ax$, and p is a prime number with $p \nmid a$ because $1 \leq a < p$, it follows that $p \mid x$. The only integer x in the range $0 \leq x < p$ satisfying $p \mid x$ is $x = 0$. Therefore, the congruence $ax \equiv 0 \pmod{p}$ has exactly one solution, which is $x = 0$.

For statement (b), suppose there exist $x_1, x_2 \in \mathbb{Z}_p \setminus \{0\}$ such that $ax_1 \equiv ax_2 \pmod{p}$, which is equivalent to $a(x_1 - x_2) \equiv 0 \pmod{p}$. From statement (a), we know that $(x_1 - x_2) \equiv 0 \pmod{p}$, which means $x_1 = x_2$. Hence, all elements $(ax \bmod p)$ are distinct for $x \in \mathbb{Z}_p \setminus \{0\}$. Since there are $(p-1)$ such elements, we conclude that $\{ax \bmod p : x \in \mathbb{Z}_p \setminus \{0\}\} = \mathbb{Z}_p \setminus \{0\}$.

For statement (c), consider the product of all the elements in $\{ax \bmod p : x \in \mathbb{Z}_p \setminus \{0\}\}$, which is $a^{p-1} \cdot (p-1)!$. On the other hand, we have derived that this set is equal to $\mathbb{Z}_p \setminus \{0\}$, and the product of all elements in $\mathbb{Z}_p \setminus \{0\}$ is $(p-1)!$. Therefore, we have $a^{p-1} \cdot (p-1)! \equiv (p-1)! \pmod{p}$. Since p is prime, $(p-1)! \not\equiv 0 \pmod{p}$, so we can divide both sides by $(p-1)!$ to obtain $a^{p-1} \equiv 1 \pmod{p}$.

For statement (d), since $\{ax \bmod p : x \in \mathbb{Z}_p \setminus \{0\}\} = \mathbb{Z}_p \setminus \{0\}$, every element in $\mathbb{Z}_p \setminus \{0\}$ can be expressed as $ax \bmod p$ for a unique $x \in \mathbb{Z}_p \setminus \{0\}$. In particular, there exists a unique $b \in \mathbb{Z}_p \setminus \{0\}$ such that $ab \equiv 1 \pmod{p}$.

Problem 14

Prove Proposition 2.54: For any two distinct integers $k, k' \in K$, if a hash function h is sampled uniformly at random from $\mathcal{H}_{p,m}$, then

$$\Pr[h(k) = h(k')] \leq \frac{1}{m}.$$

Answer:

Define event A as $h(k) = h(k')$. We ignore the case $m = 1$ since the probability is always at most 1.

For any two distinct integers $k, k' \in K$ and a fixed a, we count the number of hash functions $h_{a,b,p,m}$ that cause a collision, with b varying.

Let $x_1 = (ak + b) \bmod p$, $x_2 = (ak' + b) \bmod p$, and $d = a(k - k') \bmod p$. Thus, $x_2 = (x_1 - d) \bmod p$, which implies the collision condition is:

$$x_1 \equiv (x_1 - d) \bmod p \pmod m$$

Discussing the result of $(x_1 - d) \bmod p$:

$$(x_1 - d) \bmod p = \begin{cases} x_1 - d + p, & \text{if } 0 \leq x_1 < d, \\ x_1 - d, & \text{if } d \leq x_1 < p. \end{cases}$$

The two cases correspond to the conditions $d \equiv p \pmod m$ and $d \equiv 0 \pmod m$, respectively.

Since d is a constant, the number of b values that lead to a collision is:

$$\begin{cases} p - d, & \text{if } d \equiv 0 \pmod m, \\ d, & \text{if } d \equiv p \pmod m, \\ 0, & \text{otherwise.} \end{cases}$$

Summing over all $a \in \mathbb{Z}_p \setminus \{0\}$, we obtain the total number of collisions. As established in Problem 13, since $k \neq k'$, the value of d takes every integer in $\mathbb{Z}_p \setminus \{0\}$ exactly once.

Finally, let $p = t \cdot m + r$, where $0 \leq r < m$. The number of collision functions is given by:

$$\begin{aligned} |A| &= \sum_{i=1}^{t} (p - im) + \sum_{i=0}^{t-1} (im + r) \\ &= t(p + r - m). \end{aligned}$$

Since $m \cdot t \leq p$ and $r < m$, we can conclude that:

$$\begin{aligned} \Pr[A] &= \frac{|A|}{|\Omega|} \\ &= \frac{t(p + r - m)}{p(p - 1)} \\ &\leq \frac{1}{m}. \end{aligned}$$